Heat-engine efficiency

antocciphysics

UNIT 5 / DAY 9 / CPA

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate useful output and rejected energy for a heat engine.

SUCCESS CRITERIA - I can...

1. Use efficiency=work output/heat input.
2. Account for rejected heat.
3. Distinguish an ideal upper bound from actual performance.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

At the same kelvin temperature, sample B contains 3 times as many particles of the same ideal gas as A. Compare average energy per particle and total translational energy.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\eta = W_{out}/Q_{hot}$; $Q_{hot} = W_{out} + Q_{cold}$
$\eta_{max} = 1 - T_{cold}/T_{hot}$.
Cyclic heat engine; heat amounts are positive magnitudes. The reversible upper bound uses kelvin temperatures.

Worked reasoning: A heat engine takes 100 J from a hot source, does 25 J of work and rejects 75 J. Efficiency=25/100=25%.

Your turn: Draw three labeled energy arrows and check conservation.

Draw and label all energy transfers across the boundary.

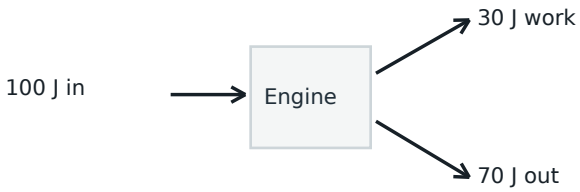
Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

An engine receives 100 J, produces 30 J of work and rejects 70 J. Check the energy balance and find efficiency.



Marzano check - circle: ① ② ③

04 Apply the physics

A model heat engine receives 200 J from a hot source per cycle and produces 50 J of useful work. Complete the flow model. Find efficiency and the heat rejected each cycle.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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05 Explain your reasoning

Why should a 95% efficient electric motor not be confused with a 95% heat engine converting heat from one source?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed mass, speed becomes 1/4 of its original value. Compare kinetic energy and the energy-equivalent TNT amount.
Show the energy changes and transfers; label units.

Initial **Transfers** **Final**



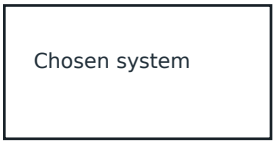
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A device outputs 40 J work from 100 J heat. State efficiency and rejected heat. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.



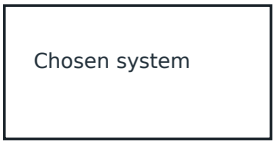
Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

An engine claims 100 J input, 40 J useful work and 50 J rejected heat. Identify the missing accounting and give one possible explanation. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3